

2. Comparison with Specialized Models

		MAE/MSE ↓				
Methods	task	Temperature	Dewpoint	Wind Direc. (×10 ³)	Wind Rate	Sea-Level Pressure
iTransformer	forecast 48h->72h	3.29/21.98	3.54/25.91	77.83/9.64	1.70/6.17	5.18/58.49
Corrformer		3/17.72	3.16/20.88	76.56/9.74	1.50/4.44	4.30/40.91
Ours	imputation	1.62/6.94	1.62/7.43	66.8/8.86	1.35/4.35	1.17/4.06
SSSD		5.49/35.21	4.51/28.23	116.82/16.46	2.52/8.85	5.67/47.54
Ours		0.91/1.80	0.90/2.12	49.0/6.01	1.00/2.09	0.55/0.61
Diffusion-TS	generation	4.23/33.06	3.06/20.44	88.58/13.59	1.94/7.80	4.22/38.68
Ours		1.79/7.27	1.82/7.93	71.9/9.50	1.43/4.47	1.27/4.57
		F1-SEDI 99.5/90 ↑				
	task	Temperature	Dewpoint	Wind Direc.	Wind Rate	Sea-Level Pressure
iTransformer	forecast 48h->72h	0.15/0.52	0.10/0.45	\	0.01/0.10	0.13/0.37
Corrformer		0.17/0.56	0.28/0.51	\	0.04/0.12	0.15/0.40
Ours	imputation	0.50/0.79	0.44/0.75	\	0.24/0.43	0.76/0.86
SSSD		0.22/0.51	0.26/0.59	\	0.17/0.27	0.31/0.57
Ours		0.74/0.89	0.69/0.86	\	0.26/0.53	0.92/0.95
Diffusion-TS	generation	0.13/0.42	0.19/0.54	\	0.15/0.25	0.21/0.49
Ours		0.45/0.76	0.36/0.71	\	0.14/0.33	0.75/0.85

3. Parameter Sensitivity Analysis

UinGIO Version	param	flops	MAE/MSE ↓					
			Temperature	Dewpoint	Wind Direc. (×10 ³)	Wind Rate	Sea-Level Pressure	
8 expert	3.73 MB	136.18 GFLOPs	1.58/5.82	1.58/6.33	66.98/8.75	1.33/3.82	1.12/3.53	
16 expert	4.01 MB	136.26 GFLOPs	1.59/5.86	1.58/6.30	69.32/8.73	1.33/3.85	1.10/3.49	
32 expert	4.57 MB	137.53 GFLOPs	1.53/5.61	1.56/6.25	67.72/8.69	1.33/3.78	1.04/3.24	
1M	0.99 MB	39.19 GFLOPs	1.60/5.79	1.61/6.36	66.97/8.59	1.34/3.80	1.16/3.61	
15M	14.63 MB	501.90 GFLOPs	1.50/5.39	1.54/6.09	66.31/8.65	1.31/3.74	1.01/3.02	
UinGIO Version	param	flops	F1-SEDI 99.5/90 ↑					
			Temperature	Dewpoint	Wind Direc.	Wind Rate	Sea-Level Pressure	
8 expert	3.73 MB	136.18 GFLOPs	0.54/0.80	0.46/0.75	\	0.17/0.39	0.79/0.88	
16 expert	4.01 MB	136.26 GFLOPs	0.52/0.79	0.45/0.74	\	0.18/0.39	0.78/0.88	
32 expert	4.57 MB	137.53 GFLOPs	0.55/0.80	0.46/0.75	\	0.19/0.39	0.79/0.88	
1M	0.99 MB	39.19 GFLOPs	0.52/0.79	0.45/0.74	\	0.17/0.38	0.75/0.87	
15M	14.63 MB	501.90 GFLOPs	0.55/0.80	0.47/0.75	\	0.21/0.42	0.80/0.89	
UinGIO Version	param	flops	FID ↓					
			Temperature	Dewpoint	Wind Direc.	Wind Rate	Sea-Level Pressure	
8 expert	3.73 MB	136.18 GFLOPs	0.24	0.25	0.57	0.39	0.17	
16 expert	4.01 MB	136.26 GFLOPs	0.24	0.25	0.56	0.38	0.17	
32 expert	4.57 MB	137.53 GFLOPs	0.23	0.24	0.55	0.37	0.15	
1M	0.99 MB	39.19 GFLOPs	0.25	0.27	0.58	0.44	0.19	
15M	14.63 MB	501.90 GFLOPs	0.22	0.24	0.53	0.33	0.14	
UinGIO Version	param	flops	STE ↓					CroS ↓
			Temperature	Dewpoint	Wind Direc. (×10 ³)	Wind Rate	Sea-Level Pressure	
8 expert	3.73 MB	136.18 GFLOPs	2.58	1.51	2.53	1.04	0.85	1.49
16 expert	4.01 MB	136.26 GFLOPs	2.57	1.5	2.51	1.03	0.84	1.49
32 expert	4.57 MB	137.53 GFLOPs	2.55	1.49	2.47	1	0.82	1.47
1M	0.99 MB	39.19 GFLOPs	2.7	1.54	2.61	1.09	0.9	1.5
15M	14.63 MB	501.90 GFLOPs	2.5	1.45	2.36	0.97	0.76	1.44